

Odd

BITS, Pilani – K K Birla Goa Campus		
CS F213, MAC F212 Object Oriented Programming	Sem 1 2025-26 MidSemester test	25% (50 Marks) October 11, 2025

INSTRUCTIONS: (1) Ensure that you are seated in the seat assigned and have the corresponding Even or Odd paper. (2) **There are 3** Quiz (10 minute each) 4:10, 4:25 and 4:50 PM (3) 2 Programming Quiz questions 5:05-5:25 and 5:10-5:30 PM (4) For the Programming Questions-**Compiler can be used from 5:05 PM.** (5) Programming questions all Class access specifiers should be **package** and methods and fields should be **public** (6) **The answers are autoevaluated- No extra spaces/characters should be entered anywhere in an answer.**

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Programming Questions

PQ4 Create a `Rectangle` class that should have **int** attributes named `centreX`, `centerY`, `len` and `width` and **color** and a `PaintSetup` object `ps`

1. The `Rectangle` constructor takes five **int** arguments `centreX`, `centerY`, `len` and `width` and **color**.
2. The `Rectangle` class has a static inner class called `PaintSetup`.
3. `PaintSetup` has a constructor which takes three **int** arguments `len`, `width` and `Color`. It sets an **int** attribute named `colorPay`. The formula is: `colorPay = (len * width)*color`
4. The `Rectangle` class has a method called `paintCalculation(int len and int width and int color)`.
5. `paintCalculation` creates an object of the `PaintSetup` class and sets it to `ps`.

Test Case

Rectangle object `r1` is constructed with argument (100, 100, 10, 10, 2)

When `r1` method `PaintCalculation(10, 10, 2)` is called it creates a `PaintSetup` object with 10, 10, 2 as constructor arguments and sets `ps` variable to point to it.

```
System.out.println(r1.ps.colorPay)
```

Output

200

PQ5. Create a `Rectangle` class that should have **int** attributes named `centreX`, `centerY`, `len` and `width` and **color**.

1. The `Rectangle` constructor takes five **int** arguments `centreX`, `centerY`, `len` and `width` and **color**.
 2. The `Rectangle` class implements the `Comparable` interface.
 3. The `Rectangle` class will be compared using `area = len * width`.
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Test Case

Rectangle object `r1` is constructed with arguments `(100, 100, 10, 10, 2)`, `r2` with `(100,100,20,20,2)`

When `r1` is compared to `r2` the output is `-1`.

When `r2` is compared to `r1` the output is `+1`

When `r1` is compared to `r1` output is `0`